



Exception Handling Problem Bank (Java)

Legend

★ = Core Placement Problem (Mandatory)

These problems should be completed by every student before moving to Streams & Lambda.

Section 1: Exception Handling Basics

- ★ Basic try-catch Block
- ★ try-catch-finally Block
- ★ Multiple catch Blocks for Different Exceptions
- ★ Handle `ArithmeticException` (Divide by Zero)
- ★ Handle `ArrayIndexOutOfBoundsException`
 - Handle `NullPointerException`
 - Handle `NumberFormatException`

Section 2: Exception Hierarchy (Checked vs Unchecked)

- ★ Difference between Checked and Unchecked Exceptions
- ★ `Throwable`, `Exception`, and `Error` Class Hierarchy Demonstration
- ★ Checked Exception Example using `IOException`
 - Unchecked Exception Example using `RuntimeException`
 - Error vs Exception Awareness (`StackOverflowError`, `OutOfMemoryError`)

Section 3: Multi-Catch and Nested Try-Catch

- ★ Multi-Catch Block using the `|` Operator
- ★ Nested try-catch Blocks
 - Rethrowing an Exception
- ★ Exception Propagation through Method Calls

Section 4: Throwing Exceptions

- ★ Throw a Custom Exception using `throw`
- ★ Declare Checked Exceptions using `throws`
- ★ Validate Input and Throw `IllegalArgumentException`
 - Method that Declares Multiple Exception Types

Section 5: Custom Exceptions (Java-Specific Emphasis)

- ★ Create a Custom Checked Exception



- ★ **Create a Custom Unchecked Exception**
- ★ **Custom Exception with a Custom Message and Constructor**
 - Custom Exception Chaining using a Cause

Section 6: Try-With-Resources (Java-Specific)

- ★ **Try-With-Resources for Automatic Resource Closing**
- ★ **Implement AutoCloseable in a Custom Class**
 - Multiple Resources in a Single Try-With-Resources Block
- ★ **Compare Try-With-Resources vs Manual finally Block Cleanup**

Section 7: Common Built-In Exceptions

- ★ **ClassCastException Demonstration**
 - NegativeArraySizeException Demonstration
- ★ **StackOverflowError from Infinite Recursion**
- ★ **ConcurrentModificationException while Modifying a Collection during Iteration**

Section 8: Real World Exception Handling Applications

- ★ **Bank Withdrawal System with Custom InsufficientFundsException**
- ★ **File Reading System with Proper Exception Handling**
 - User Input Validator with Custom Exceptions
- ★ **ATM PIN Validation with Retry Limit and Custom Exception**
 - Age Validation System using Custom Exception